

View Reviews

Paper ID

72

Paper Title

StatLM: A Hybrid Pipeline for Zero-shot Vietnamese Multi-Label Text Classification

Track Name

Main Session - KSE 2026

Reviewer #4

Questions

1. Comments to Authors

The paper introduces StatLM, a hybrid encoder-decoder pipeline for open-world zero-shot multi-label text classification, specifically tailored for Vietnamese text. The system uses a statistical encoder to filter and condense raw documents before passing them to a large language model (LLM). Layer 1 uses embedding-based TextRank to summarize the text, Layer 2 uses KeyBERT with an asymmetric retrieval model to extract keyphrases, and Layer 3 uses Gemini 2.5 Flash to generate, normalize, and assign coarse-grained labels based only on those keyphrases.

Strengths

- Good efficiency gains: Compared to the MiniLM (X-MLClass) full-LLM baseline, StatLM reduces average processing time from 40 seconds to 10 seconds per document. It requires approximately 30 times fewer tokens and cuts API calls from eight down to one per document.
- High label quality: By shielding the LLM from raw-text noise, StatLM increases the coarse-label ratio from 32.6% to 76.9% and shrinks the noise-label ratio from 39.1% to just 3.8%.
- High coverage: The method successfully generates labels that cover 19 of the 20 editorial categories in the balanced 2,000-article VietnamNet corpus.

Limitations:

The paper does not compare StatLM against direct LLM classification using advanced prompting techniques (like Chain-of-Thought or iterative self-refinement) on the raw text, which might mitigate the noise issues of direct LLM approaches without requiring a multi-stage statistical encoder.

3. Overall Score

Weak Accept

Reviewer #5

Questions

1. Comments to Authors

Summary:

This paper proposes StatLM, a hybrid pipeline for open-world zero-shot multi-label Vietnamese text classification. It condenses documents using embedding-based TextRank and asymmetric KeyBERT, then invokes an LLM solely to discover, normalize, and assign coarse labels. Evaluated on a 2,000-article VietnamNet corpus, StatLM outperforms an X-MLClass baseline in label quality, category coverage, processing latency, and token efficiency, supported by thorough component-level and ablation analyses.

Strengths:

The paper proposes an intuitive, cost-effective hybrid pipeline that restricts LLM usage to the final stage, yielding notable operational savings in processing time and token consumption while improving coarse-label quality. The experimental evaluation is thorough and well-documented, systematically benchmarking each component, computational overhead, and ablation configuration.

Weaknesses:

In the abstract and introduction, the baseline is introduced as an X-MLClass-based full-LLM pipeline, but it is suddenly labeled as "MiniLM" in Table V and throughout the discussion. This creates unnecessary confusion for the reader. The authors should keep the naming consistent by using "X-MLClass", or "X-MLClass (Gemini)" if they wish to highlight the Gemini backbone used.

The keyphrase ground truth for Layer 2 is generated by another LLM (DeepSeek V4) without human verification, raising concerns about potential evaluation bias. Adding a human-verified sample would make this evaluation more solid. Additionally, using a single editorial category as a proxy may not fully capture the framework's multi-label capabilities, and testing on true multi-label data would be beneficial.

3. Overall Score

Weak Accept

Reviewer #7

Questions

1. Comments to Authors

Summary of paper:

The paper proposes StatLM, a hybrid pipeline for zero-shot, "open-world" multi-label classification of Vietnamese documents. StatLM uses two statistical encoder layers, embedding-based TextRank for summarization and KeyBERT for keyphrase extraction, before using an LLM decoder to discover, normalize, and assign labels. The paper also introduces a 2000-article VietnamNet corpus as the main dataset used for benchmarking.

Comments:

1. My main concern is that the evaluation does not have a true ground-truth label space. The paper explicitly states that the original 20 VietnamNet categories are used as a "coarse-label proxy" because no absolute label-space ground truth exists.

The proposed task is "open-world" zero-shot multi-label classification, while the evaluation effectively evaluates whether generated labels can recover a known 20-category taxonomy. This creates a mismatch between the claims and the evaluation.

2. The keyphrase references are LLM-generated from DeepSeek V4. This weakens the reliability of the keyphrase evaluation because the proposed method is itself designed to use an LLM downstream.

3. Overall Score

Borderline